On Completeness of SDP-Based Barrier Certificate Synthesis over Unbounded Domains

Hao Wu [1] Shenghua Feng [2] Ting Gan [3]
Jie Wang [4] Bican Xia [5] Naijun Zhan [1, 5]

[1]Institute of Software, Chinese Academy of Sciences (CAS)
[2]Zhongguancun Laboratory
[3]Wuhan University
[4]Academy of Mathematics and Systems Science, CAS
[5]Peking University

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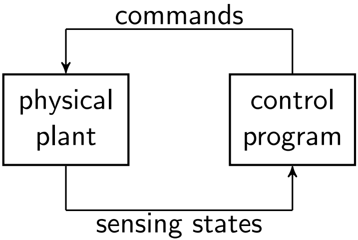
Embedded Systems



- In **safety critical** scenarios, we want to ensure the physical plant will not evolve into unsafe states.
- **Safety verification problem** for continuous-time dynamical systems.

ODE Systems

System dynamics given by Ordinary Differential Equations (ODEs):

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x})$$

- $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^n$ is a polynomial vector field,
- Unique trajectory $\xi_{\boldsymbol{x}_0} : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ for any initial state $\boldsymbol{x}_0$.

Safety Verification Problem of ODE Systems

Safety Verification Problem: Given domain $\mathcal{X} \subseteq \mathbb{R}^n$, initial set $\mathcal{I} \subset \mathcal{X}$, and unsafe set $\mathcal{U} \subset \mathcal{X}$ such that $\mathcal{U} \cap \mathcal{I} = \emptyset$. Prove that

$\mathcal{U}$ is not reachable from any state in $\mathcal{I}$.

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- One of the key challenges in verification, undecidable in general (Henzinger et al., 1995).
- Method 1: compute/approximate the reachable set.
- Method 2: find a differential invariant. ✓

Differential Invariants

- Analogous to **inductive invariants** for programs (discrete-time systems).
- A **differential/positive invariant** is a subset $\Phi \subseteq \mathcal{X}$ such that any trajectory starting from Φ stays within Φ forever, i.e.,

$$\forall \boldsymbol{x}_0 \in \Phi, \forall t \in \mathbb{R}_{\geq 0}.\ \xi_{\boldsymbol{x}_0}(t) \in \Phi.$$

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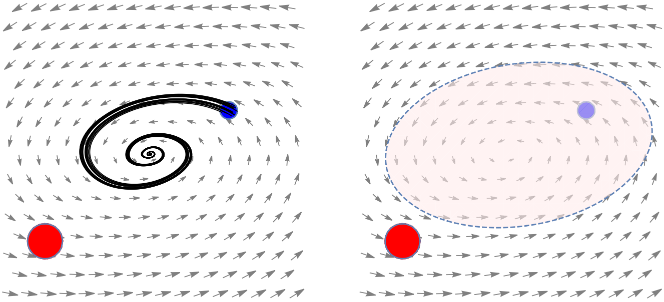

Barrier Certificates (BCs)

- A real-valued function $B : \mathbb{R}^n \to \mathbb{R}$ is a **barrier certificate** if the set $\{\boldsymbol{x} \in \mathbb{R}^n \mid B(\boldsymbol{x}) \leq 0\}$ is a differential invariant.

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- For simplicity, we consider exponential-type BC conditions modified from Kong et al., 2013:

$$\begin{aligned} &\forall \boldsymbol{x} \in \mathcal{I}.\ B(\boldsymbol{x}) \le 0, \\ &\forall \boldsymbol{x} \in \mathcal{U}.\ B(\boldsymbol{x}) \ge \epsilon_e, \\ &\forall \boldsymbol{x} \in \mathcal{X}.\ \mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) - \lambda B(\boldsymbol{x}) \le 0, \end{aligned}$$

where
  - $\mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) \ \hat{=}\ \langle \frac{\partial}{\partial \boldsymbol{x}} B(\boldsymbol{x}), \boldsymbol{f}(\boldsymbol{x}) \rangle$ is the Lie derivative.
  - $\lambda \in \mathbb{R}$ is a user-defined constant, we set $\lambda = -1$.
  - $\epsilon_e > 0$ is a constant (to avoid $\mathcal{I}$ and $\mathcal{U}$ being arbitrarily close).

SDP-based Synthesis: SOS polynomials

- A polynomial $\sigma(\boldsymbol{x})$ is **Sum-of-Squares** (SOS) if it can written as

$$\sigma(\boldsymbol{x}) = \sum_{i=1}^{n} p_i^2(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}].$$

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- $\Sigma[\boldsymbol{x}]$ denotes the set of SOS polynomials.
- A polynomial $f(\boldsymbol{x})$ is SOS iff the Gram matrix of f is PSD.

$$\begin{aligned} f(x_1, x_2) &= 2x_1^4 + 2x_1^3 x_2 - x_1^2 x_2^2 + 5x_2^4 \\ &= \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ x_1 x_2 \end{pmatrix}'}_{\text{basis}} \underbrace{\begin{pmatrix} 2 & -3 & 1 \\ -3 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix}}_{\text{Gram matrix}} \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ x_1 x_2 \end{pmatrix}}_{\text{basis}} \\ &= \frac{1}{2}(2x_1^2 - 3x_2^3 + x_1 x_2)^2 + \frac{1}{2}(x_2^2 + 3x_1 x_2)^2 \end{aligned}$$

SDP-based Synthesis: Putinar's Positivstellensatz

- Let $S = \{x \mid p_i(x) \geq 0,\ i = 1, \ldots, m\}$.
- **Quadratic Module**

$$\mathbf{QM}(S) \triangleq \{\sigma_0 + \sum_{i=1}^{m} \sigma_i p_i \mid \sigma_i \in \Sigma[\boldsymbol{x}] \text{ for } 1 \leq i \leq m\}$$

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- A quadratic module **QM** is **Archimedean** if $N - \|\boldsymbol{x}\|^2 \in \mathbf{QM}$ for some constant $N \in \mathbb{N}$.
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Theorem (Putinar's Positivstellensatz)

Assume the Archimedean condition holds. If $f \in \mathbb{R}[x]$ is strictly positive on S, then $f \in \mathbf{QM}(S)$, i.e.,

$$f(x) = \sigma_0(x) + \sum_{i=1}^{m} p_i(x) \cdot \sigma_i(x), \quad \sigma_i(x) \in \Sigma[x].$$

SDP-based Synthesis: Translation

- Assume $\mathcal{X}$,$\mathcal{I}$,$\mathcal{U}$ are basic closed semialgebraic sets.
- Consider $B(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}]$, translate BC conditions into SOS constraints:

$$\begin{cases} \forall \boldsymbol{x} \in \mathcal{I}.\ B(\boldsymbol{x}) \leq 0, \\ \forall \boldsymbol{x} \in \mathcal{U}.\ B(\boldsymbol{x}) \geq \epsilon_e, \\ \forall \boldsymbol{x} \in \mathcal{X}.\ \mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) - \lambda B(\boldsymbol{x}) \leq 0. \end{cases} \to \begin{cases} -B(\boldsymbol{x}) \in \mathbf{QM}(\mathcal{I}), \\ B(\boldsymbol{x}) - \epsilon_e \in \mathbf{QM}(\mathcal{U}) \\ \lambda B(\boldsymbol{x}) - \mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) \in \mathbf{QM}(\mathcal{X}) \end{cases}$$

Under some mild conditions (**finite convergence property**)

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- Question: How to obtain completeness over unbounded domains?

Algebraic Tool: Homogenization

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- Let $p(\boldsymbol{x})$ be a polynomial of degree d, its **homogenization** is

$$\tilde{p}(\tilde{\boldsymbol{x}}) \ \hat{=} \ x_0^d \cdot p(\frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0}).$$

$$p(x_1, x_2) = x_1^2 + x_2 + 1 \implies \tilde{p}(x_0, x_1, x_2) = x_1^2 + x_2 x_0 + x_0^2.$$

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- Let $S = \{\boldsymbol{x} \in \mathbb{R}^n \mid p_1(\boldsymbol{x}) \geq 0, \ldots, p_m(\boldsymbol{x}) \geq 0\}$, define

$$\tilde{S} \ \hat{=} \ \{\tilde{\boldsymbol{x}} \in \mathbb{R}^{n+1} \mid \tilde{p_1}(\tilde{\boldsymbol{x}}) \geq \ 0, \ldots, \tilde{p}_m(\tilde{\boldsymbol{x}}) \geq \ 0, \ \underbrace{x_0 \geq \ 0, \|\tilde{\boldsymbol{x}}\|^2 = 1}_{\text{half unit sphere in } \mathbb{R}^{n+1}} \}.$$

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- An one-to-one projection between S and $\tilde{S}_+$:

$$\boldsymbol{x} \in S \iff \left(\frac{1}{\sqrt{1+\|\boldsymbol{x}\|^2}}, \frac{x_1}{\sqrt{1+\|\boldsymbol{x}\|^2}}, \ldots, \frac{x_n}{\sqrt{1+\|\boldsymbol{x}\|^2}}\right) \in \tilde{S}_+.$$

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- Points with $x_0 = 0$ in $\tilde{S}$ correspond to **points at infinity** in $\mathbb{R}^n$.
- For these points, a technical assumption is needed.

Algebraic Tool: Homogenization

Definition (closed at infinity (Nie, 2012))

A basic semialgebraic set S is closed at ∞ if

$$cl(\tilde{S}_+) = \tilde{S}.$$

- Being closure at infinity is a generic property for semialgebraic sets.
- Depends on description polynomials.

$$S_1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 - x_2^2 \geq 0\} \text{ not closed at } \infty$$
$$S_2 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 - x_2^2 \geq 0, \underbrace{x_1 \geq 0}_{\text{redundant}}\} \text{ closed at } \infty$$

Algebraic Tool: Homogenization

Theorem (Equivalence (Huang et al., 2023))

When S is closed at ∞,

$$f(\boldsymbol{x}) \geq 0 \text{ over } S \iff \tilde{f}(\tilde{\boldsymbol{x}}) \geq 0 \text{ over } \tilde{S}$$

- The quadratic module $\mathbf{QM}(\tilde{S})$ is Archimedean.

Constraints for Polynomial BC

- Assume that $\mathcal{I}$, $\mathcal{U}$, and $\mathcal{X}$ are all closed at infinity.
- First **homogenization**, then **Putinar's Positivstellensatz**.

$$\begin{cases} -\tilde{B}(x_0, \boldsymbol{x}) = \textbf{QM}(\tilde{\mathcal{I}}) \\ \tilde{B}(x_0, \boldsymbol{x}) - \epsilon_e x_0^d = \textbf{QM}(\tilde{\mathcal{U}}) \\ \tilde{H}(x_0, \boldsymbol{x}) = \textbf{QM}(\tilde{\mathcal{X}}) \end{cases}$$

where $H(\boldsymbol{x}) = \lambda B(\boldsymbol{x}) - \mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x})$.

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- Under some mild conditions, both sound and complete.

From the system's perspective...

- What we have done:
  1. Project a system in $\mathbb{R}^n$ into a new bounded system over in $\mathbb{R}^{n+1}$.
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- Let $(y_0, y_1, \ldots, y_n) = (\frac{1}{\sqrt{1+\|\boldsymbol{x}\|^2}}, \frac{x_1}{\sqrt{1+\|\boldsymbol{x}\|^2}}, \ldots, \frac{x_n}{\sqrt{1+\|\boldsymbol{x}\|^2}})$ be the state vector of the new system.

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Theorem

$B(y_0, \boldsymbol{y}) \in \mathcal{C}^1(\mathbb{R}^{n+1})$ is a barrier certificate of the homogenized system if and only if $B(\frac{1}{\sqrt{\|\boldsymbol{x}\|^2+1}}, \frac{\boldsymbol{x}}{\sqrt{\|\boldsymbol{x}\|^2+1}})$ is a barrier certificate of the original system.

Semialgebraic Barrier Certificates

- To avoid rational functions, we simplified by

$$(\sqrt{\|\boldsymbol{x}\|^2+1})^d \cdot B(\frac{1}{\sqrt{\|\boldsymbol{x}\|^2+1}}, \frac{\boldsymbol{x}}{\sqrt{\|\boldsymbol{x}\|^2+1}})$$
$$= \underbrace{B_1(\boldsymbol{x}) + \sqrt{\|\boldsymbol{x}\|^2+1} \cdot B_2(\boldsymbol{x})}_{\textbf{semialgebraic BC}}$$

where $B_1, B_2 \in \mathbb{R}[\boldsymbol{x}]$.

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- Semialgebraic BC subsumes polynomial BC as a special case, where $B_2(\boldsymbol{x})$ is constant zero.
- Semialgebraic BC can also be synthesized using SDP.

Experiments

			Incomplete			Polynomial BC			Semialgebraic BC		
system	dim	unbounded	deg	succ	time(s)	deg	succ	time(s)	deg	succ	time(s)
vector[37]	2	$\mathcal{X}$	4	✓	0.58	3	✓	0.03	**2**	✓	0.83
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.39	4	✓	0.16	**2**	✓	0.72
barrier[29]	2	$\mathcal{X}$	>6	✗	2.40	>6	✗	2.73	>4	✗	22.98
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.78	3	✓	0.04	**2**	✓	3.12
lie-der[24]	2	$\mathcal{X}$	3	✓	0.19	3	✓	0.14	**1**	✓	0.75
		$\mathcal{I},\mathcal{U},\mathcal{X}$	**3**	✓	0.14	**3**	✓	0.19	**3**	✓	5.04
arch1[36]	2	$\mathcal{X}$	**4**	✓	0.40	**4**	✓	0.54	>4	✗	117.61
		$\mathcal{I},\mathcal{U},\mathcal{X}$	**1**	✓	0.09	**1**	✓	0.04	2	✓	20.92
arch2[36]	2	$\mathcal{X}$	**3**	✓	0.20	**3**	✓	0.21	**3**	✓	5.53
		$\mathcal{I},\mathcal{U},\mathcal{X}$	3	✓	0.18	3	✓	0.18	**2**	✓	1.59
arch3[36]	2	$\mathcal{X}$	**2**	✓	0.12	**2**	✓	0.13	**2**	✓	3.45
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.84	>6	✗	1.30	**1**	✓	0.34
arch4[36]	2	$\mathcal{X}$	>6	✗	0.11	5	✓	0.74	**3**	✓	5.69
		$\mathcal{U},\mathcal{X}$	>6	✗	1.02	**6**	✓	1.21	>4	✗	7.74
nagumo[34]	2	$\mathcal{X}$	**2**	✓	0.13	**2**	✓	0.14	**2**	✓	3.50
		$\mathcal{U},\mathcal{X}$	>6	✗	1.02	**3**	✓	0.17	>4	✗	23.73
lorenz[10]	3	$\mathcal{X}$	6	?	TO	**4**	✓	72.13	2	?	TO
		$\mathcal{U},\mathcal{X}$	**5**	✓	248.88	6	?	TO	2	?	TO
lotka[14]	3	$\mathcal{X}$	>6	✗	88.97	>6	✗	27.8	3	?	TO
		$\mathcal{U},\mathcal{X}$	>6	✗	0.27	>6	✗	438.54	3	?	TO

Experiments

			Incomplete			Polynomial BC			Semialgebraic BC		
system	dim	unbounded	deg	succ	time(s)	deg	succ	time(s)	deg	succ	time(s)
vector[37]	2	$\mathcal{X}$	4	✓	0.58	3	✓	0.03	**2**	✓	0.83
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.39	4	✓	0.16	**2**	✓	0.72
barrier[29]	2	$\mathcal{X}$	>6	✗	2.40	>6	✗	2.73	>4	✗	22.98
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.78	3	✓	0.04	**2**	✓	3.12
lie-der[24]	2	$\mathcal{X}$	3	✓	0.19	3	✓	0.14	**1**	✓	0.75
		$\mathcal{I},\mathcal{U},\mathcal{X}$	**3**	✓	0.14	**3**	✓	0.19	**3**	✓	5.04
arch1[36]	2	$\mathcal{X}$	**4**	✓	0.40	**4**	✓	0.54	>4	✗	117.61
		$\mathcal{I},\mathcal{U},\mathcal{X}$	**1**	✓	0.09	**1**	✓	0.04	2	✓	20.92
arch2[36]	2	$\mathcal{X}$	**3**	✓	0.20	**3**	✓	0.21	**3**	✓	5.53
		$\mathcal{I},\mathcal{U},\mathcal{X}$	3	✓	0.18	3	✓	0.18	**2**	✓	1.59
arch3[36]	2	$\mathcal{X}$	**2**	✓	0.12	**2**	✓	0.13	**2**	✓	3.45
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.84	>6	✗	1.30	**1**	✓	0.34
arch4[36]	2	$\mathcal{X}$	>6	✗	0.11	5	✓	0.74	**3**	✓	5.69
		$\mathcal{U},\mathcal{X}$	>6	✗	1.02	**6**	✓	1.21	>4	✗	7.74
nagumo[34]	2	$\mathcal{X}$	**2**	✓	0.13	**2**	✓	0.14	**2**	✓	3.50
		$\mathcal{U},\mathcal{X}$	>6	✗	1.02	**3**	✓	0.17	>4	✗	23.73
lorenz[10]	3	$\mathcal{X}$	6	?	TO	**4**	✓	72.13	2	?	TO
		$\mathcal{U},\mathcal{X}$	**5**	✓	248.88	6	?	TO	2	?	TO
lotka[14]	3	$\mathcal{X}$	>6	✗	88.97	>6	✗	27.8	3	?	TO
		$\mathcal{U},\mathcal{X}$	>6	✗	0.27	>6	✗	438.54	3	?	TO

- Expressiveness: Semialgebraic $>$ Polynomial $>$ Incomplete
- Efficiency: Incomplete $\approx$ Polynomial $\gg$ Semialgebraic

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- We also applied this technique to **Craig interpolation synthesis problem**, another paper at FM2024.

Thanks! & Questions?

Henzinger, Thomas A., Peter W. Kopke, Anuj Puri, and Pravin Varaiya (1995). "What's decidable about hybrid automata?" In: *Proceedings of the Twenty-Seventh Annual ACM Symposium on Theory of Computing, STOC 1995*. ACM, pp. 373–382.

Huang, Lei, Jiawang Nie, and Ya-Xiang Yuan (2023). "Homogenization for polynomial optimization with unbounded sets". In: *Mathematical Programming* 200.1, pp. 105–145.

Kong, Hui, Fei He, Xiaoyu Song, William N. N. Hung, and Ming Gu (2013). "Exponential-Condition-Based Barrier Certificate Generation for Safety Verification of Hybrid Systems". In: *Computer Aided Verification, 25th International Conference*. Vol. 8044. Lecture Notes in Computer Science. Springer, pp. 242–257.

Nie, Jiawang (2012). "Discriminants and nonnegative polynomials". In: *Journal of Symbolic Computation* 47.2, pp. 167–191.